

Timothy Tarter

Harrisonburg, VA | 757-608-1555 | tartertx@dukes.jmu.edu | <https://github.com/Tartertx>

EDUCATION

James Madison University

B.S. Mathematics

Minors: Computational Analytics

Anticipated Graduation: December 2026

Cumulative GPA: 3.6

Dean's List: Fall 2023, 2024, 2025, Spring 2024, Summer 2024

PROJECTS & PRESENTATIONS

Conference Talk: Gannon University's Applied Intelligence Conference & Joint Mathematics Meetings 2026

Title: "A Faster Attack on LFSR Recurrences: an Alternative to the Berlekamp-Massey Algorithm"

Invited Symposium Address: James Madison University, Department of Mathematics

Title: "On Y-Delta and X-Square Graph Reductions and the Study of their Algebraic Properties"

Invited Conference Publication: Graduate Society for International Studies Conference 2025

Title: "Modeling the Impact of NATO Policy Actions on Online Threat Populations via the Lotka-Volterra Framework"

Link: <https://arxiv.org/abs/2508.01904>

Conference Paper: Shenandoah Undergraduate Mathematics Symposium Fall 2024 & Joint Mathematics Meetings 2025

Title: "Applications of Benford's Law Generalized to Geospatial Threats and Social Network Investigations"

Link: <https://arxiv.org/abs/2501.10460>

PROFESSIONAL EXPERIENCE

Capital Markets and Index Solutions Intern, Annexus

May 2025 – August 2025

- Researched proprietary mean-reversion and trend-following strategies for trading and index construction.
- Utilized the Pandas, Numpy, and Matplotlib packages in Python and Excel to implement backtests.

Teaching Assistant, James Madison University

August 2024 – May 2025

- Selected by professor recommendation to serve as a TA for Multivariable Calculus (Fall 2024) and Real Analysis (Spring 2025) by holding weekly office hours, hosting test prep seminars, and teaching material to students.
- Wrote practice exams and solutions guides in LaTeX / Overleaf.

Mathematician, JMU Football

September 2024 – May 2025

- Developed econometric models for player performance which impacted player lineups, starting rosters, and scholarships for the 2025 season.
- Met weekly with chief decisionmakers to provide progress updates and wrote a complete research report.

Mathematician, Hacking for Defense

August 2024 – December 2024

- Led the applied mathematics team in developing mathematical models for White Sands Missile Range to interpolate drone flight paths from extremely limited detection data using methods in measure theory, convex optimization, and stochastic optimization.
- Interview & Success Story: <https://www.h4d.us/student-outcomes/timothy-tarter>

LEADERSHIP EXPERIENCE & RESEARCH GRANTS

Brakke Mason Research Endowment

April 2025 - Present

- Awarded \$1,900 to research Bruhat Order on the symmetric via the study of Schubert Calculus on F_2^n .

President, Lecturer, & Lead Quantitative Researcher, Madison Institute for Mathematical Finance

May 2025 – Present

- Lead the Computational Algebraic Geometry (CAG) research group in developing statistical arbitrage extraction algorithms. Currently researching the use of twisted cohomology on integer modules for recovering forcing functions from snapshots of integer density plots as a proponent of "Snapshot Learning."
- Developed and taught informal coursework in Calculus, Linear Algebra, Differential Equations, Numerical Analysis, Probability Theory, Real Analysis, Measure Theory, Stochastic Differential Equations, and Financial Derivatives. Also designed a class covering advanced topics in Group Theory, Galois Theory, Classical Cryptography for CAG members.
- Worked through chapters 1-5 of Vakil's 'The Rising Sea: Foundations of Algebraic Geometry' (Lecture recordings are available at <https://www.youtube.com/@TimothyTarter>). (Continued work is in progress)
- Manage all research teams and coordinate trade implementation on behalf of the Quantitative Trading group.

President, JMU Math Club

August 2024 – May 2025

- Organized academic symposiums, problem solving events, fundraisers, and social mixers.

COMPUTATIONAL SKILLS

LaTeX – Expert

Python – Proficient in Pandas, NumPy, sympy & matplotlib packages

R Studio – Experienced in econometric modeling

Microsoft Excel – Experienced in backtesting & financial mathematics

Maucaalay2 – Some experience with Gröbner basis computation

MATLAB – General experience